SLIIT UNT
THE KNOWLEDGE UNIVERSITY

IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-073

1. Topic (12 words max)

An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-073
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka's education-to-employment ecosystem faces a critical disconnect between students' academic choices, employability outcomes, and long-term wellbeing. Each year, thousands of students complete the G.C.E. Advanced Level (A/L) examination without access to structured, data-driven guidance on suitable degree pathways or alternative career options. National and international studies highlight persistent skills mismatches and weak alignment between education outputs and labor market needs in Sri Lanka [1]. Conventional career guidance mechanisms rely heavily on static aptitude tests and subjective counseling, offering limited personalization and minimal transparency into decision-making [2]. This often results in students enrolling in misaligned degree programs, experiencing academic stress, inefficient study patterns, dropping out, or graduating with qualifications that do not align with labor market demand.

Existing digital career guidance platforms predominantly focus on either academic counseling or job matching in isolation. They rarely integrate academic performance, skill development pathways, salary expectations, and mental wellbeing into a unified decision-support system. Furthermore, many AI-based recommender systems function as "black boxes," providing recommendations without explainable reasoning, which undermines trust in high-stakes educational and career decisions [3][4]. Equally important, current systems fail to recognize student time and cognitive capacity as finite resources, often leading to unrealistic workload planning, burnout, and reduced learning effectiveness issues well documented in cognitive load theory and self- regulated learning research [5][6]. These challenges are particularly pronounced in the Sri Lankan context due to socio-economic constraints, regional disparities, and rapidly evolving skill requirements. This research proposes an explainable, AI-driven career intelligence platform tailored to Sri Lanka's socio-economic and educational landscape. The system integrates academic pathway prediction, skills-to-job matching, context-aware workload balancing, and verified mentorship into a single ecosystem. Using explainable artificial intelligence (XAI) techniques, the platform not only recommends suitable degrees or job tracks but also provides human-readable explanations and counterfactual insights [3][4]. Its wellbeing component further converts study effort into measurable competency progress and dynamically adapts task difficulty based on the user's productivity flow, drawing on principles from behavioral learning analytics and adaptive learning systems [5][7][8]. By aligning academic potential, market demand, and sustainable workload management, the proposed solution aims to reduce educational mismatch, improve employability, and support long-term career development.

References

1. World Bank. (2020). Sri Lanka – Education and Skills Development. World Bank Group. Supports the education–employment mismatch and skills gap context in Sri Lanka.

2. OECD. (2021). Career Guidance for Adult Learners in a Changing World of Work. OECD Publishing. Justifies the need for integrated, lifelong, and data-driven career guidance systems.

3. Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. arXiv preprint arXiv:1702.08608.Foundational reference for explainable AI (XAI) and transparent decision-support systems.

4. Kaur, H., et al. (2020). Interpretable and explainable AI systems for human decision-making. Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems.Supports the importance of explainability in high-stakes human-centered AI applications.

5. Winne, P. H., & Hadwin, A. F. (2008). The weave of motivation and self-regulated learning. Motivation and Self-Regulated Learning: Theory, Research, and Applications.Grounds the competency-based and self-regulated learning framework.

6. Sweller, J., Ayres, P., & Kalyuga, S. (2011). Cognitive Load Theory. Springer. Directly supports treating time and cognitive capacity as finite resources.

7. Baker, R. S., & Inventado, P. S. (2014). Educational data mining and learning analytics. In Learning Analytics (pp. 61–75). Springer. Supports behavioral learning analytics and productivity-aware adaptation.

8. Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.Provides theoretical grounding for behavioral momentum and adaptive task difficulty.

6. Brief description of Existing Research and Systems (300 words max)

This project proposes an AI-Driven Career Intelligence and Decision-Support Platform designed to assist Sri Lankan students in making informed academic and career decisions. Unlike traditional career guidance methods that rely on generic advice or one-time counseling sessions, this platform continuously analyzes student data, academic performance, personal interests, and local labor-market trends to provide personalized, data-driven career insights. The system integrates four core components. The Academic Pathway Predictor uses machine learning models trained on historical university performance data, subject combinations, and progression trends to forecast suitable degree pathways and potential academic outcomes. This helps students identify optimal subject selections and reduce the risk of dropouts or misaligned academic choices.

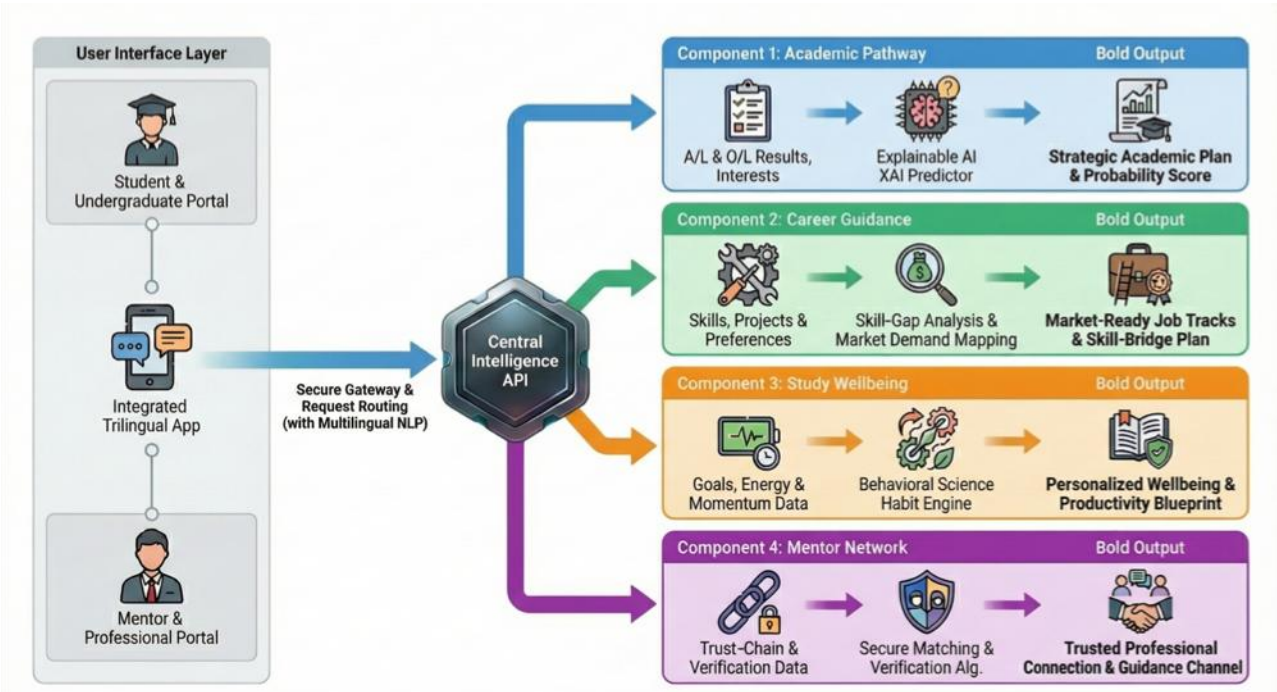
The Career Guidance and Job-Match Recommender aligns student skills, qualifications, and interests with real-time job market demand in Sri Lanka. Using AI-based recommendation algorithms and external job-market datasets, the platform suggests career paths, emerging roles, and required skill gaps, enabling students to plan proactively.

The Study Wellbeing and Time Management Assistant monitors study patterns, workload distribution, and self-reported stress levels to provide adaptive scheduling suggestions, burnout alerts, and productivity guidance. This component promotes balanced academic performance and mental wellbeing.

The Professional Mentorship and Alumni Network connects students with verified industry professionals and alumni through intelligent matching based on career goals and academic background. Structured mentorship interactions support real-world exposure and career readiness.

A centralized AI engine processes data from all components and delivers insights through a user-friendly web or mobile application. Conceptually, student inputs and external datasets feed into the AI analytics layer, which outputs personalized recommendations, visual dashboards, and actionable guidance. This platform bridges the gap between education and employment while supporting sustainable career development for Sri Lankan students.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).



The proposed solution is an integrated, AI-driven career intelligence platform designed to support Sri Lankan students from post-A/L decision-making through employability and sustainable academic wellbeing. The system is architected as a modular, service-oriented platform that combines explainable artificial intelligence, skills-based career matching, context-aware workload management, and human mentorship into a unified decision-support ecosystem. As illustrated in the conceptual diagram, users interact with the system through a trilingual web and mobile interface (Sinhala, Tamil, and English), accessible via a Student & Undergraduate Portal and a Mentor & Professional Portal. All user requests are securely routed through a Central Intelligence API, which acts as the orchestration layer. This gateway handles authentication, data validation, multilingual natural language processing, and intelligent routing of requests to the relevant functional components.

Component 1: Academic Pathway Intelligence focuses on post-A/L academic decision-making. Students submit their O/L and A/L results, academic history, and personal interests. An explainable AI (XAI) predictor processes these inputs to generate ranked degree pathway recommendations with associated probability scores. Unlike black-box models, the system provides transparent, human-readable explanations and counterfactual insights, enabling students to understand why specific academic paths are suggested and what improvements could increase success likelihood. The output is a strategic academic plan supported by evidence-based reasoning.

Component 2: Career Guidance and Job-Match Recommender supports undergraduates and non-degree-seeking students by translating skills, projects, and preferences into realistic job pathways. This module performs skill-gap analysis and maps user profiles to market-ready job tracks based on curated labor-market data. Outputs include suitable entry-level roles, indicative salary ranges, identified skill gaps, and recommended micro-credentials, forming a structured skill-bridge plan aligned with market demand.

Component 3: Study Wellbeing and Productivity Intelligence introduces a novel, behavior-aware layer that treats student time and cognitive capacity asfinite resources. Using goals, energy signals, and momentum data, a behavioral science habit engine dynamically generates competency-linked micro-actions. Study effort is translated into measurable skill progression rather than raw hours, while task difficulty and workload are adaptively adjusted based on the learner's current productivity flow. The output is a personalized wellbeing and productivity blueprint that balances academic ambition with sustainable workload management.

Component 4: Mentor and Professional Network provides the human guidance layer of the system. Using trust-chain verification data, the platform securely matches students with verified university seniors or industry professionals. A privacy-preserving verification algorithm ensures mentor credibility without storing sensitive credentials. The output is a trusted professional connection and guidance channel that complements AI-driven recommendations with real-world insights. Overall, the solution functions as a context-aware, explainable, and ethically designed career intelligence ecosystem that integrates data-driven guidance, behavioral adaptation, and human mentorship to support informed, sustainable education-to-employment transitions.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful development of the proposed AI-driven career intelligence platform is supported by the availability of relevant domain expertise, technical knowledge, and accessible data sources. The project draws upon established academic, industry, and technological foundations in the areas of education systems, career guidance, artificial intelligence, behavioral science, and human-centered system design,ensuring both feasibility and academic rigor.Domain Expertise in education and career guidance is well established within the Sri Lankan context. University academics, school counselors, and higher-education administrators possess deep knowledge of the G.C.E. O/L and A/L examination structures, university admission criteria, degree pathways, and common student performance trends. This expertise enables accurate modeling of academic progression patterns and validation of degree suitability rules. Additionally, industry professionals and HR practitioners contribute practical insights into employability requirements, entry-level job roles, and skill expectations across key sectors such as IT, engineering, finance, healthcare, and technical services, supporting realistic skills-to-job mapping.

Technical Expertise in artificial intelligence, machine learning, and data engineering is widely available through local universities, research institutions, and the growing Sri Lankan software industry. Expertise in supervised learning, probabilistic modeling, recommender systems, and explainable AI (XAI) techniques such as SHAP and LIME supports the development of transparent prediction and recommendation models. Furthermore, expertise in learning analytics, behavioral modeling, and adaptive systems enables the implementation of context-aware workload balancing and behavioral-momentum mechanisms that dynamically adjust task difficulty, duration, and sequencing based on user productivity patterns. Full-stack development skills including web and mobile UI design, API development, multilingual natural language processing, and secure system architecture are readily accessible, enabling end-to-end system implementation and deployment.

Data Availability is sufficient for training and validating the proposed models using ethically sourced and consent-based datasets. Academic performance data (O/L and A/L results, subject streams, and grade distributions) are standardized and publicly documented by the Department of Examinations, Sri Lanka.Historical university intake statistics, degree structures, and progression outcomes are available through the University Grants Commission (UGC) and institutional reports.

Labor market information, including job role definitions, skill requirements, and approximate salary ranges, can be curated from government publications, industry surveys, and verified recruitment platforms, avoiding the need for live data scraping.

Knowledge Resources related to wellbeing, study habits, and behavioral science are supported by extensive literature in educational psychology, cognitive load theory, habit formation, and self-regulated learning. These resources inform the design of the wellbeing-aware study and time management component, particularly the treatment of student time as a finite resource and the conversion of study effort into measurable competency progress. Mentorship system design is guided by existing research on peer learning, alumni engagement, and trust-based digital communities.

Overall, the convergence of local educational expertise, established AI and behavioral science knowledge, and accessible structured datasets provides a strong foundation for the proposed research.This availability significantly reduces implementation risk while enabling the development of a scalable, context-aware, and ethically responsible career intelligence platform.

9. Objectives and Novelty

Main Objective To design and develop an integrated, AI-driven career intelligence ecosystem tailored to the Sri Lankan context that supports students and young adults in making informed, realistic, and wellbeing-aware academic and career decisions by intelligently linking academic performance, skills, financial and market realities, behavioral support, and human mentorship into a single explainable decision-support platform.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Rusiru L.W.S IT22220288	Academic Pathway Predictor	• Collect and structure student academic data (A/L results, O/L history, interests, and aptitude responses) required for pathway prediction. • Design and implement a probabilistic prediction model to	• Explainable AI (XAI) focused on degree selection for after-A/L students, not just classification but human-readable counterfactuals. • Singlish preprocessing so students can type naturally in Sinhala/Tamil/English mixes and still get accurate inputs.

		estimate student success likelihood across multiple degree programs. • Analyze historical academic outcome patterns to map student profiles with suitable academic pathways. • Integrate Explainable AI (XAI) techniques to generate clear, human-readable explanations for each degree recommendation. • Develop counterfactual reasoning logic to provide "if-then" improvement guidance (e.g., grade improvement impact on eligibility). • Generate personalized action plans, including recommended bridging courses and preparatory study steps.	

		• Design and implement a trilingual (Sinhala/Tamil/English) user interface with Singlish-aware input preprocessing. • Develop backend APIs for prediction, explanation delivery, and action-plan generation. • Test and validate model accuracy, explanation clarity, and usability with sample student scenarios	
Wickremathilaka J.A IT22152114	Adaptive Career Pathway and Skills-Gap Recommender System	• Collect user inputs: current degree, declared skills, projects/internships, preference, and career interests. • Process skills ontology and taxonomy to map declared skills against job role requirements.	• The component introduces a skills-gap mapping system that identifies which abilities a student currently lacks for a chosen career path. It then provides a structured plan to close those gaps using short courses or micro-credentials. • It is tailored specifically for Sri Lankan undergraduates and A/L students who are not pursuing further academic studies,

		• Run recommender engine to match degree + skills to 1–2 suitable job tracks. • Generate skill-gap report highlighting missing skills and readiness timeline. • Curate salary ranges from validated datasets with source/date/confidence metadata. • Recommend micro-credentials (2–3 short courses) to close identified gaps. • Provide career guidance output: role description, salary range, common tasks, and learning plan. • Deliver frontend components: skill-gap dashboard, job recommendation cards, timeline	making the guidance locally relevant and practical. • Instead of scraping live salary data, the system uses a curated salary dataset. Each salary range is presented with the date of collection and a confidence level, ensuring the information is reliable, feasible to maintain, and legally safe.

		planner, and multilingual guidance text. • Maintain backend services: recommender API, curated course list, salary dataset.	
Mahikalpa K.B.V.I IT22150370	Professional Mentorship & Alumni Network	• Design mentor–student matching logic based on academic interests, degree programs, and career goals. • Implement automated mentor verification using university email/domain or simulated API validation. • Develop a secure real-time communication channel (chat) between students and mentors. • Create mentor profiles with verified badges and controlled visibility settings.	• Automated trust-chain verification eliminating manual mentor validation. • Privacy-preserving credibility checks without credential storage. • Trusted Mentor badge system to improve confidence in guidance quality. • Integration of human mentorship as a formal decision-support layer, not an external add-on.

		• Ensure privacy-preserving verification without storing sensitive credentials. • Integrate the mentorship module with the central intelligence API for seamless system interaction.	
	Intelligent Assistant for Academic Productivity and Mental Wellbeing	• Study Pattern Monitoring Track daily and weekly study hours Identify peak productivity times and low-focus periods • Workload Distribution Analysis Analyze assignment deadlines, exams, and study load Detect uneven workload distribution across days/weeks • Stress Level Collection Allow students to self-report stress, fatigue, and motivation levels	• Wellbeing-Aware Time Management Unlike traditional planners, the system integrates mental wellbeing indicators directly into scheduling decisions. • Adaptive & Context-Aware Scheduling Study plans automatically adapt based on real-time workload and stress levels, not fixed timetables. • Early Burnout Prevention Detects burnout risks before performance drops, enabling preventive intervention rather than reactive support.

		Combine subjective stress data with study behavior • Adaptive Scheduling Generate personalized study schedules based on workload and stress Adjust plans dynamically when stress or overload is detected • Burnout Detection & Alerts Identify early signs of burnout (long study hours + high stress) Notify students with rest, break, or rescheduling recommendations • Productivity Guidance Suggest optimal study durations and break intervals Provide tips for focus, rest, and recovery • Wellbeing Insights Dashboard Visualize study balance, stress trends, and productivity scores Help	• Balanced Productivity Model Focuses on sustainable academic performance, balancing efficiency with mental health instead of maximizing study hours. • Personalized Student Experience Each student receives recommendations tailored to their habits, stress tolerance, and academic demands.

		students reflect on habits and improve long-term wellbeing	

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Ms	Thilini	Jayalath	
Co-Supervisor	Ms.	Jenny	Krishara	Jenny 2026/01/13
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				